

Chapter one

Introduction to Medical Mycology

Objectives

➤ At the end of this chapter, students will be able to:

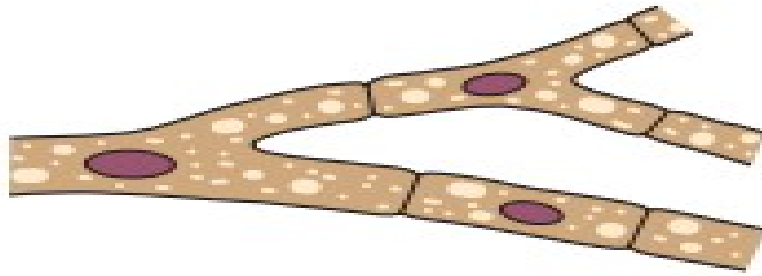
- ❑ Define Medical Mycology
- ❑ Describe characteristics of fungi
- ❑ List the differentiating characteristics of fungi from bacteria.
- ❑ Explain the morphology of fungi.
- ❑ Discuss the disease of fungi.

Introduction to Medical Mycology

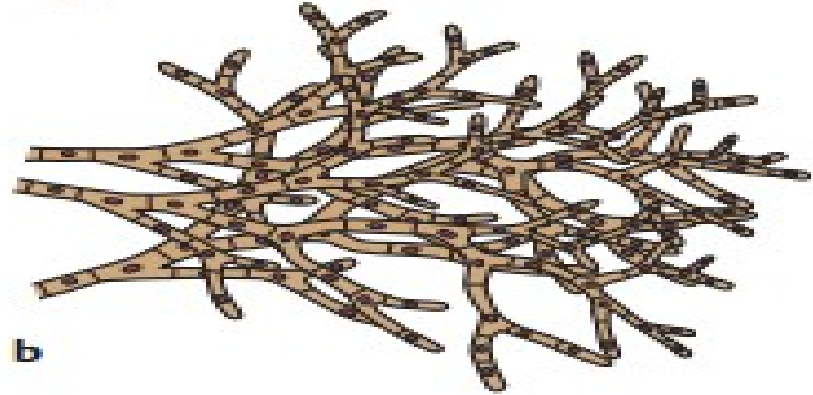
- Mycology is the study of **fungi(fungus)**
- Medical mycology: fungi that cause disease in humans.
- **Fungus** are:
 - Eukaryotic organisms, their cells contain membrane-bound organelles and clearly defined nuclei
 - Rigid cell walls, which possess two characteristic cell structures: **chitin** and **ergosterol**
 - Lack the definite root, stem, and leaves of highly organized plants.
 - Include macroscopic and microscopic forms.
 - Are among the most widely distributed organisms on Earth
 - Most are saprobes(live on dead organisms)

- Fungi are clearly distinguished from all other living organisms including animals, by their principal modes of **vegetative growth** and **nutrient intake**.
 - **grow** from the **tips of filaments** (hyphae): **hyphae** (singular hypha), **mycelium** and **thallus**
 - Digest organic matter **externally** before absorbing it into their mycelia.
- Between 100,000 to 200,000 species known, out of which about **300** associated with human diseases
- Two basic morphological(**Dimorphism**); fungi grow as **yeasts** in vivo, as well as in vitro at 37°C and as **molds** at 25°C
 - regulated by factors such as
 - temperature,
 - CO₂ concentration,
 - pH,
 - levels of cysteine or other sulfhydryl-containing compounds

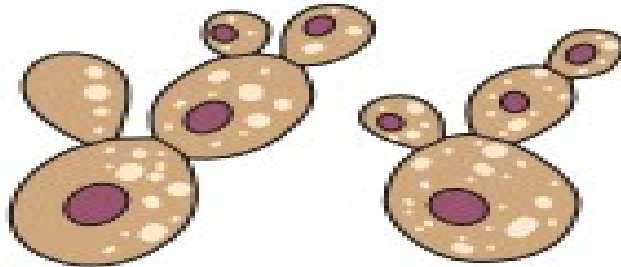
Basic Morphological Elements of Fungi



a



b



c



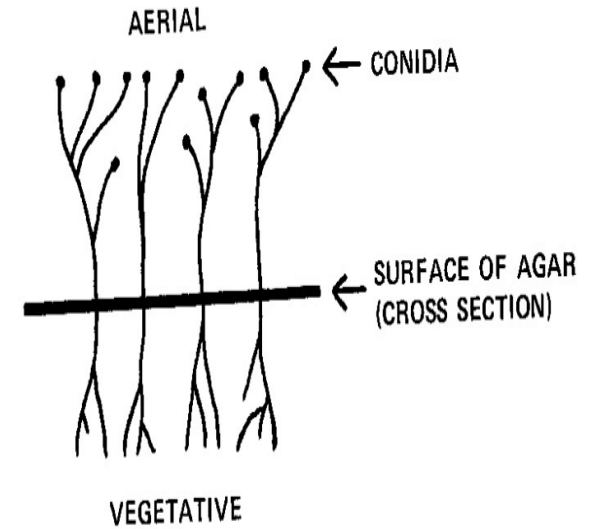
d

Fig. 5.1 There are two basic morphological forms: hypha and yeast.

A Hypha, septate, or nonseptate. **B** Mycelium: web of branched hyphae. **C** Yeast form, budding. **D** Pseudohyphae.

Mycelium

- **Mass of branching intertwined hyphae**
 - Vegetative
 - Aerial
- **Vegetative mycelium**
 - Nutrient absorbing and water exchanging portion of the fungi
 - Grows in or on a substrate.
 - Are actively growing and form the main body of the colony.
- **Aerial hyphae,**
 - Fungal filaments or hyphae that protrude above the surface
 - Produce asexual spore
 - responsible for reproduction
- **Eucarpic and Holocarpic thallus**

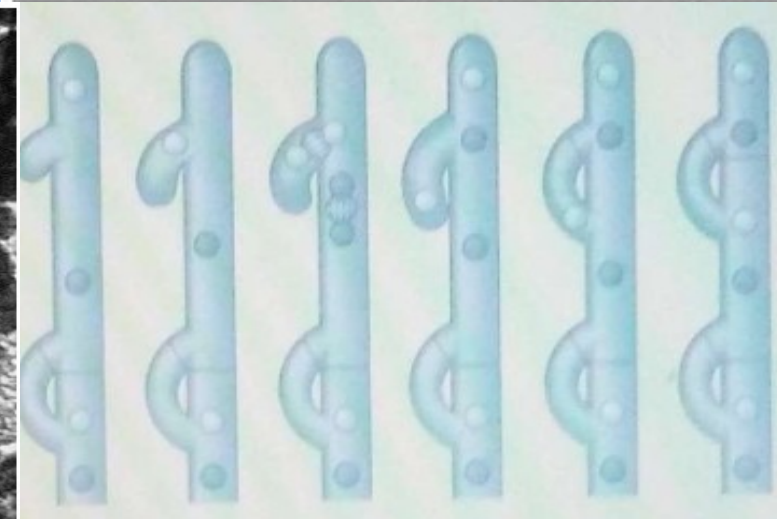
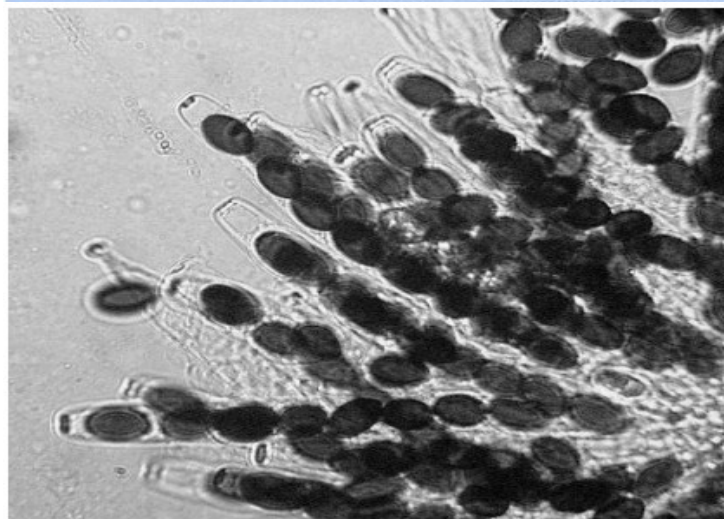
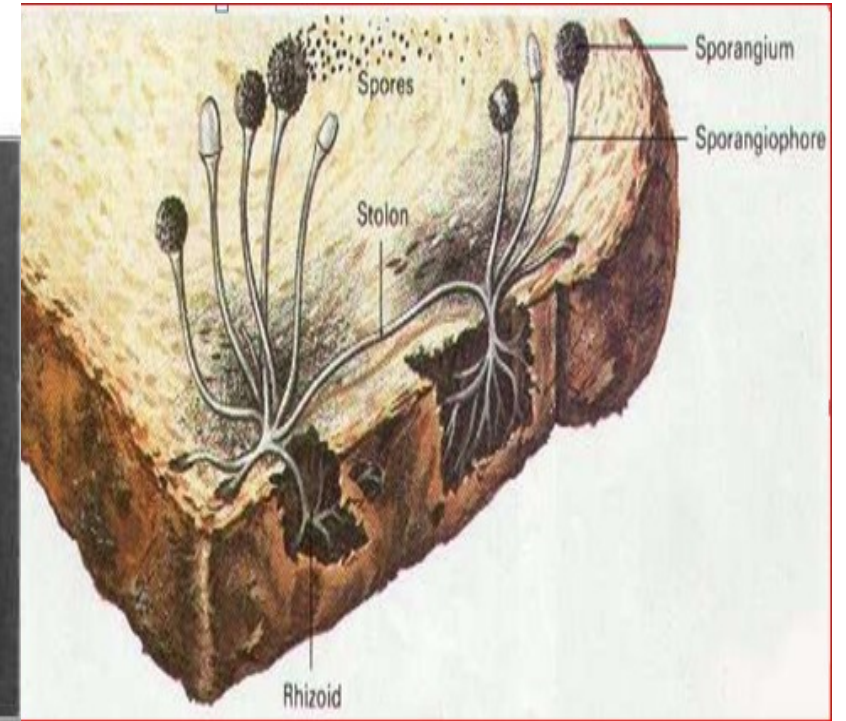
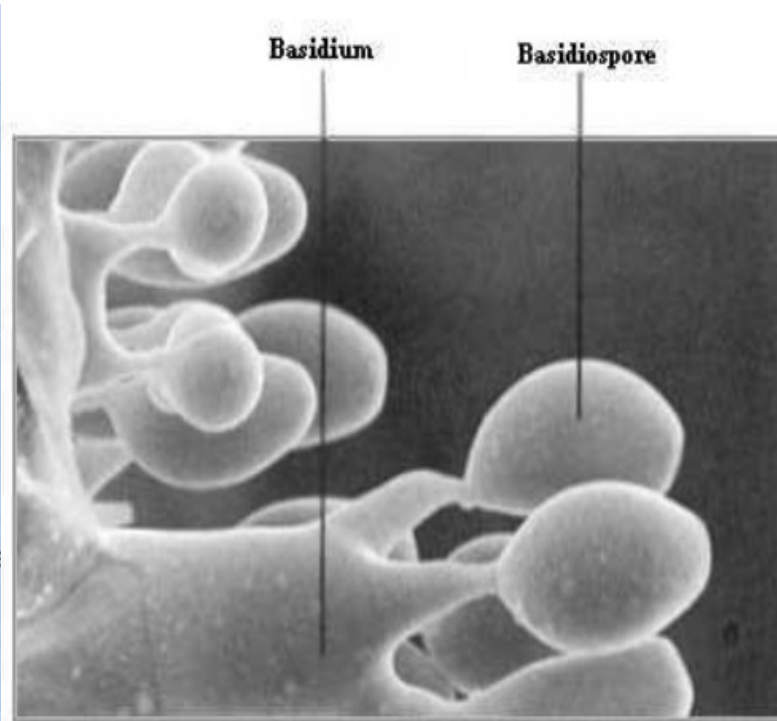
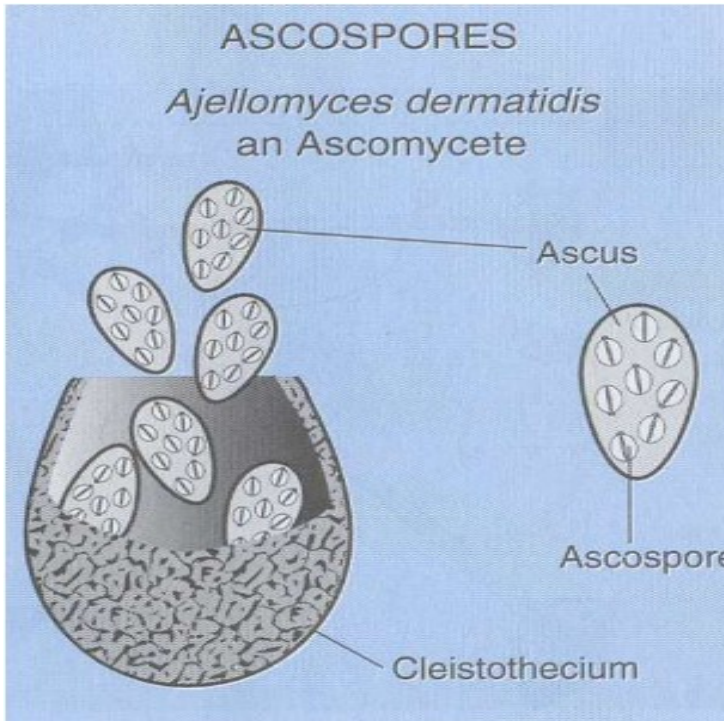


Difference between Fungi and Bacteria

Properties	Fungi	Bacteria
Nucleus	Eukaryotic; nuclear membrane; more than one chromosome; mitosis	Prokaryotic; no membrane; nucleoid; only one “chromosome”
Cytoplasm	Mitochondria; endoplasmic reticulum; 80S ribosomes	No mitochondria; no endoplasmic reticulum; 70S ribosomes
Cytoplasmic membrane	Sterols (ergosterol)	No sterols
Cell wall	Glucans, mannans, chitin, chitosan	Murein, teichoic acids (Gram-positive), proteins
Metabolism	Heterotrophic; mostly aerobes; no photosynthesis	Heterotrophic; obligate aerobes and anaerobes, facultative anaerobes
Size, mean diameter	Yeast cells: 3–5–10 μm . Molds: indefinable	1–5 μm
Dimorphism	In some species	None

Classification of Fungi

- There are two types(basically)
 - Biological classification
 - Sexual reproduction, morphology(structure of cross wall or septa, life cycle, and physiology)
 - Site of pathology
- The kingdom Fungi is currently divided into four phyla or divisions based mainly on their **sexual reproduction**
 - **Ascomycota** **ascospores**
 - Basidiomycota **basidiospores**
 - Zygomycota **zygospores**
 - Chytridiomycota **Zoospores**



Deuteromycetes

- ❑ Also known as the “imperfect fungi.”
- ❑ Species with no sexual reproduction identified yet are classified under this phyla
- ❑ Many of these species have recently been reclassified into one of the other phyla when the sexual stage was discovered.
- ❑ Most Deuteromycota have a well-developed, septate mycelium with distinct conidiophores but some have a unicellular thallus.
- ❑ Deuteromycetes occur mostly as saprophytes on a wide range of substrates
 - but a large number of them are parasites on plants and animals (including humans) and cause a variety of diseases.

Significance of Fungi

Clinical importance

- a. Infections (mycoses)
 - Superficial
 - Cutaneous
 - Subcutaneous
 - Systemic
- b. **Mycotoxicosis**
 - toxic secondary metabolites
- c. **Hypersensitivity reactions**
 - Fungal allergens
- d. **Mycetismus**
 - ingestion of preformed toxin/mushroom poisoning

Importance in nature

- energy cycle, as decomposers; composting
 - biogeochemical cycling of elements in nature
 - detoxification of organic pollutants and for bioremediation of heavy metals
- food (e.g. edible mushrooms) and food production: enzymes and organic acids
 - East Asian foods: *A. oryzae* and *A. sojae*
- well-established model organisms in cell biology and genetics
 - *A. nidulans*
- pharmaceuticals
 - echinocandins: *A. nidulans* and *A. sydowi*
 - lovastatin: *A. terreus*; fumagillin: *A. fumigatus*

- **The interest in fungal infections has increased, because:-**

- ↑ frequency of infection

- ↑ Awareness of clinicians

- ↑ invasive procedures

- ↑ Use of immuno-suppressive drugs

- ↑ Immuno-suppressed individuals

Fungal infection

- Fungi cause some of the most persistent and disfiguring disease still prevalent throughout the world.
- Diseases caused by fungi are collectively called mycoses (singular, mycosis).
- The incidence of the disease (infection) is related to the degree of exposure to fungi in living conditions, occupation, and leisure activities and to immune status.
- There are five types of mycoses in two main categories:
 - **Skin mycoses**
 - **Systemic mycoses**

The clinical nomenclatures used for the mycoses are based on the

A. Site of the infection

- There are four general categories on the basis of the primary tissue affinity of the pathogen.
 - ✓ Superficial (**outer layers of the skin and hair**)
 - ✓ Cutaneous(**deeper in the epidermis, hair and nails**)
 - ✓ Subcutaneous(**dermis, subcutaneous tissues and muscle**)
 - ✓ Systemic mycoses

B. Route of acquisition of the pathogen

- ✓ Exogenous (airborne, cutaneous)
- ✓ Endogenous (colonization by a member of the normal flora or reactivation of a previous infection).

C. Type of pathogenicity exhibited by the fungus.

- ✓ Primary pathogens can establish infections in normal hosts.
- ✓ Opportunistic pathogens cause disease in individuals with compromised host defense mechanisms.

Superficial mycoses

- ❑ Fungal infections of the skin and hair that invade only the most **superficial layers** and cause little or no inflammatory response.
- ❑ The fungi produce **keratinase**, which allows them to metabolize and live on human keratin tissue like **skin** and **hair**.
- ❑ Are a major cause of morbidity in the world, particularly in the **tropics**, where **heat** and **humidity** provide the ideal conditions for the growth of fungi .
- ❑ **Direct contact** is sufficient to **transmit** the infection from a contaminated surface or host to another.
- ❑ Include the following fungal infections and their etiological agent:
 - Black piedra (*pedraia hortae*)
 - White piedra (*trichosporon beigeli*)
 - Pityriasis versicolor (*malassezia furfur*)
 - Tinea nigra (*phaeoannellomyces werneckii*)

Laboratory diagnosis

- **Specimen**, Skin scrapings or hairs with respective nodules
- **Direct Examination**, 10% KOH
 - Septate hyphal elements, **asci**, and **ascospores** are seen
- **Isolation and Culture**
 - grows on sabourad dextrose agar (SDA)
 - colonies are cream colored

